SSL Certificate Tool

User Guide

Windows & macOS All-in-One Package

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This guide explains how to install and use the SSL Certificate Tool without installing Python. Use it to convert customer certificates into Ameyo-ready files (JKS, P12, PEM, Asterisk, ca-bundle) and optionally deploy them over SSH.

1. Install (No Python Required)

1.1 Windows

- Unzip SSLCertTool-Windows.zip to any folder (example: Desktop\SSLCertTool-Windows).
- Keep the whole folder together. Do not move only the .exe file.
- Double-click: Start SSL Certificate Tool.bat (or SSLCertTool.exe).
- Optional CLI: Start CLI Wizard.bat or ssl-auto.exe --wizard

1.2 macOS

- Unzip SSLCertTool-Mac.zip.
- First launch: right-click Start SSL Certificate Tool.command -> Open -> Open.
- Later: double-click the same shortcut or SSLCertTool.app.
- If blocked: System Settings -> Privacy & Security -> Allow Anyway.
- Optional fix: xattr -dr com.apple.quarantine SSLCertTool-Mac

1.3 Package contents

File / Folder	Purpose
SSLCertTool.exe / SSLCertTool	Desktop GUI
ssl-auto.exe / ssl-auto	Command-line tool
configs/	YAML deploy configs (editable)
output/	Generated certificate files
INSTALL.txt / README.txt	Quick reference

2. What You Need Before Starting

- Customer certificate (EML email, ZIP, CRT/PEM, folder, etc.).
- Matching private key (.key) for that certificate.
- Keystore password and alias used on the server (for JKS/P12).
- Optional: server paths / YAML config for SSH deploy.

Supported inputs: .crt .cer .pem .p7b .p12 .pfx .ca-bundle .eml .txt .docx .zip .7z and folders.

3. How to Use the GUI (Recommended)

3.1 Generate certificates

- Open SSLCertTool.
- Go to the Generate tab.
- Add Certificate Files (EML / folder / CRT / ZIP).
- Select Private Key (.key).
- Set Keystore Password (must match Tomcat keystorePass).
- Set Key Alias (usually ameyo-server).
- Choose Output Directory.
- Click Generate.
- Confirm Key/Cert match: PASS.

3.2 Important password rule

If the server uses connector.https.keystorePass=YOUR_PASSWORD, generate JKS/P12 with that same password. Do not leave changeit unless the server also uses changeit.

3.3 Inspect / Deploy tabs

- Inspect: check leaf CN, expiry, chain, and key match before deploy.
- Deploy: uses configs/*.yaml to copy files over SSH (always dry-run first).

4. How to Use the CLI

4.1 Interactive wizard

```
ssl-auto --wizard
```

4.2 Generate

```
ssl-auto generate --input "customer_folder_or.eml" --key "domain.key" --alias ameyo-server --passwor
```

4.3 Inspect

```
ssl-auto inspect --input "customer_folder_or.eml"
```

4.4 Deploy

```
ssl-auto plan --config mycustomer
ssl-auto deploy --certs output/domain/ --config mycustomer --dry-run
ssl-auto deploy --certs output/domain/ --config mycustomer
```

4.5 Config helpers

```
ssl-auto init --name mycustomer --domain example.com
ssl-auto configs
ssl-auto validate --config mycustomer
```

5. Generated Files (What Goes Where)

Generated file	Typical use
main_{domain}.crt	CRM / Apache leaf (often crm.crt)
{domain}.key	Private key
chain.crt / crm-ca.crt	Apache chain (usually intermediate)
{domain}.ca-bundle	Legacy Ameyo ca-bundle (int + root)
{domain}.pem	Full cert chain (certs only)
asterisk.pem	Asterisk WSS: KEY first + certs
asterisk_{domain}.crt	Asterisk cert chain
{domain}.jks	Tomcat / AppServer keystore
keystore.p12	PKCS12 keystore

5.1 Asterisk PEM order (critical)

asterisk.pem must start with the private key, then leaf, then intermediate, then root (if present). If http.conf uses tlscertfile=asterisk.pem with no tlsprivatekey, a certs-only PEM will prevent port 8089 / WebRTC from starting.

```
Correct order inside asterisk.pem:
  1) BEGIN PRIVATE KEY / RSA PRIVATE KEY
  2) BEGIN CERTIFICATE  (leaf)
  3) BEGIN CERTIFICATE  (intermediate)
  4) BEGIN CERTIFICATE  (root, optional)
```

6. Typical Ameyo Deploy Mapping

Service	Copy to (example)
AppServer / Reports JKS	/dacx/ssl2025/<name>.jks
CRM cert	/dacx/ssl2025/crm.crt
CRM key	/dacx/ssl2025/<domain>.key
CRM chain / ca-bundle	/dacx/ssl2025/chain.crt or *.ca-bundle
Asterisk	.../asterisk/keys/asterisk.pem (+ .crt/.key)

Always backup old files first. After copy: verify key matches cert, restart services (Tomcat/AppServer, Apache/CRM, Asterisk). Confirm HTTPS and WSS (8089) are listening.

7. Verification Commands (on server)

7.1 Key matches certificate

```
openssl x509 -noout -modulus -in crm.crt | openssl md5
openssl rsa  -noout -modulus -in domain.key | openssl md5
# Both MD5 values must be identical
```

7.2 Certificate dates

```
openssl x509 -in crm.crt -noout -subject -dates
```

7.3 JKS alias and password

```
keytool -list -v -keystore app.jks -storepass YOUR_PASS -alias ameyo-server
```

7.4 Asterisk WSS port

```
ss -tlnp | grep 8089
echo | openssl s_client -connect HOST:8089 2>&1 | head -20
```

7.5 Asterisk PEM structure

```
grep -E "BEGIN PRIVATE KEY|BEGIN CERTIFICATE" asterisk.pem
```

8. Troubleshooting

Problem	Fix
App won't open (Mac)	Right-click Open / allow in Privacy
Key/Cert match FAIL	Wrong key; use matching private key
Tomcat SSL fails	JKS password must equal keystorePass
WebRTC / 8089 down	asterisk.pem KEY-first; restart Asterisk
Must open IP:8089 first	Cert CN != IP; use domain in WSS URL
No .jks created	Install Java keytool or use keystore.p12
ca-bundle too short	Use {domain}.ca-bundle (int+root)

9. Building the Package (IT / Maintainers)

End users do not build. Maintainers build on each OS, then share the ZIP.

```
Windows build machine:
  cd ssl-cert-tool
  build-windows.bat
  -> dist\SSLCertTool-Windows.zip

Mac build machine:
  cd ssl-cert-tool
  chmod +x build-mac.sh && ./build-mac.sh
  -> dist/SSLCertTool-Mac.zip
```

A Mac binary must be built on a Mac. Java is optional for end users unless they need JKS generation (P12/PEM work without Java).

10. Quick Checklist

[ ] Unzip package; launch GUI or CLI wizard
[ ] Load customer cert + matching private key
[ ] Set keystore password/alias to match server
[ ] Generate; confirm Key/Cert match PASS
[ ] Check asterisk.pem starts with PRIVATE KEY
[ ] Backup old server certs
[ ] Deploy/copy files using correct names
[ ] Verify modulus match + dates + ports 443/8089
[ ] Restart AppServer / CRM / Asterisk as needed
[ ] Test HTTPS and WebRTC troubleshooter

For support, keep the full package folder and the output/ directory from the generation run.